



Value-driven attentional capture prevents commission errors in real time

Shivang Shelat^{1,2} · Barry Giesbrecht^{1,2}

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Abstract

Failures of sustained attention are pervasive in everyday life. In the lab, these states of attention can be tracked unobtrusively using response times (RTs) in monotonous, continuous performance tasks requiring frequent responses. Here, we used a RT-based real-time triggering procedure to investigate the interplay between fluctuations of attentional state and another attentional phenomenon: experience-dependent capture. We investigated predictions from two related accounts. One proposes attentional lapses could reduce one's ability to suppress capture by the distractor (capture-suppression account). The other proposes capture could “snap” attention back into focus (perceptual-recoupling account). Participants first completed a training task that differentiated colored targets by reward and then performed a sustained attention task that required they execute visual search on each trial. On rare trials, the colors relevant during training served as distractors inserted during lapsing and focused states. The results aligned with predictions from the perceptual-recoupling account: Experiment 1 showed that both novel and reward-associated distractors enhanced accuracy, whereas Experiment 2 showed a moderation of the reward-driven effect during fast-triggered moments. Pooled analyses confirmed a reliable reward-driven reduction in errors that diminished over successive blocks and was most pronounced early on for participants with stronger color–reward associations during fast-triggered trials. This study shows the potential of using experience-dependent capture to mitigate performance decrements associated with lapsing attention.

Keywords Sustained attention · Perceptual decoupling · Attentional capture · Commission errors

Introduction

During every waking moment, attention is in continual flux between optimal and suboptimal states (Esterman & Rothlein, 2019; Mooneyham & Schooler, 2013; Smallwood & Schooler, 2006; Thomson et al., 2015). In the lab, the Sustained Attention to Response Task (SART) is used to study sustained attention by requiring participants to respond repetitively to frequent stimuli while withholding/altering responses to rare trials (Robertson et al., 1997). Its monotony induces mind-wandering (Smallwood & Schooler, 2006)

and commission errors (Robertson et al., 1997). Commission errors are a common behavioral consequence of lapsing attention used in the mind-wandering and sustained-attention literatures (Chidharom et al., 2025; Christoff et al., 2009; Corriveau et al., 2024; Shelat & Giesbrecht, 2025; Smallwood et al., 2011, 2013).

Recent work has integrated the SART with real-time, closed-loop triggering to unobtrusively monitor fluctuations in cognitive state and insert events, allowing researchers to test interactions between lapses and other phenomena (for a review, see Shelat et al., 2024). This approach compares intrasubject mean response time (RT) in defined windows (i.e., successive trials) to cumulative RT means across all preceding trials. Faster windows are linked to several behavioral hallmarks of lapsing attention: task-unrelated thoughts (Marcusson-Clavertz et al., 2012; Smallwood et al., 2008b), reduced comprehension and memory encoding (deBettencourt et al., 2018, 2019; Franklin et al., 2011), and commission errors (McVay & Kane, 2012). Collectively, the evidence suggests that relative RT is a viable behavioral

✉ Shivang Shelat
sshelat@ucsb.edu

¹ Department of Psychological & Brain Sciences,
University of California, Santa Barbara, Psychology East,
Santa Barbara, CA 93117, USA

² Institute for Collaborative Biotechnologies, University
of California, Santa Barbara, Bioengineering Building,
Santa Barbara, CA 93117, USA

correlate of attentional state, and commission errors can reflect a downstream consequence of that state.

Here, we used the real-time triggering paradigm to test the relationship between lapses and another well-studied phenomenon: experience-dependent attentional capture. Experience-dependent capture refers to when a previously task-relevant stimulus feature, such as color, attracts attention even when it is task-irrelevant. This can result from selection history (e.g., blue captures attention if it previously defined a target), novelty (e.g., green captures attention if it was absent during a previous task or rare in the current task), or reward history (e.g., red captures attention if it was previously rewarded; Anderson & Yantis, 2013; Anderson et al., 2011; Awh et al., 2012). Capture is usually assessed in visual search tasks, where eye movements are drawn to distractors and/or RTs to a target are slowed by a distractor that has a feature relevant to participants' history.

In the current study, two independent samples first completed a training task where they searched for colored targets. One color was sometimes rewarded, one color unrewarded, and one color entirely absent from training. Afterward, they completed the SART with real-time triggering that required visual search for a target on each trial. Rare target trials included distractors with colors from or absent from the training session. We analyzed the data in the context of two theoretical frameworks: a *capture-suppression* account and a *perceptual-recoupling* account. We made distinct empirical predictions based on these accounts, but it is important to note that these are broadly compatible accounts because they might represent different temporal stages of the same underlying process. A more detailed explanation of the relationship between these two accounts is provided in the Discussion. The goal of our work was to assess whether the data align with our predictions from either or both accounts.

The capture-suppression account is rooted in the idea that attentional lapses represent failures of cognitive control (McVay & Kane, 2010), and since cognitive control is important in mitigating distraction (Lavie, 2010; Luck et al., 2021), suppression of capture by distractors should be weakened during lapses. Evidence for weakened capture suppression is sparse. Zhang et al. (2024) found that participants successfully inhibited initial eye movements to distractors even when they self-reported mind-wandering, but their target RTs were slower. This finding suggests that although lapsing attention does not predict greater overt saccades to a distractor, target processing during salience-driven capture is still impaired. Using a similar paradigm, Zhang et al. (2021) showed that pretrial gaze variability, a possible trial-by-trial marker of poor attention, predicted increased saccades to distractors (see also Zhang & Jonides, 2021). The discrepancy between Zhang et al. (2024) and Zhang et al. (2021) may be because self-report mind-wandering probes were not a sufficient measure of attentional engagement and

more covert, precise measures are necessary to reveal greater capture during lapses. Other work suggests that increased working memory load heightens susceptibility to value-driven capture (Watson et al., 2019), highlighting the role of executive resources in capture suppression. By this logic, on a sustained attention task with reward-associated distractors, there should be greater slowing of RTs on error trials compared to accurate trials when the reward-associated distractor is present. Further, individuals with lower working memory capacity (WMC) should exhibit larger RT costs in the presence of the reward-associated distractor during inaccurate trials. Thus, the capture-suppression account predicts that a reward-associated distractor would affect RT more on inaccurate than accurate trials, and that this effect would be moderated by WMC.

The perceptual-recoupling account posits that capture by distractors during lapses can improve performance by reengaging attention. Lapses are associated with perceptual decoupling, where focus begins to shift away from external stimuli to internally generated content (Schooler et al., 2011). Decoupling impairs stimulus processing and memory encoding (Blondé et al., 2022), and neural signatures of decoupling have been linked to errors on the SART (Smallwood et al., 2008a). High visual salience during rare SART trials recouples attention to the task, improving accuracy (Smallwood, 2013), but in a SART that demands visual search on each trial, it is uncertain whether a salient *distractor* would confer the same benefit. Accordingly, the perceptual-recoupling account predicts that a reward-associated distractor would enhance accuracy relative to a neutral distractor (in early blocks before color–reward associations are unlearned; Milner et al., 2023). On top of this, we asked whether the benefit scales with reward learning magnitude, because stronger color–reward learning should produce a stronger value-driven priority signal; and whether it is larger on fast-triggered (error-prone) trials where there may be greater room for improvement than on slow-triggered trials where attention is more engaged.

To assess the empirical support for our predictions from these two accounts, we investigated whether experience-driven capture alters behavioral performance in the SART by examining how value-, selection-, and novelty-associated distractors influenced accuracy and RTs. To test suppression, we compared RT slowing on inaccurate versus accurate trials when experience-dependent distractors were present, expecting greater slowing if suppression failed during lapse-induced errors. Note that under this framework, slowed RTs would be interpreted as a cost to performance (akin to existing visual search paradigms). To test recoupling, we assessed whether experience-dependent distractors could actually reduce those errors relative to neutral distractors. Under this framework, although we primarily focused on accuracy as the outcome variable,

a slowed RT would be interpreted as a benefit to performance as it could enable participants to respond correctly to the rare trial. Our findings showed that some distractors rescued performance during lapsing moments in the SART, supporting the predictions of the perceptual-recoupling account. In Experiment 1, both novelty-associated and reward-associated distractors enhanced rare trial accuracy, whereas Experiment 2 replicated primarily the reward-driven effect. Pooled analyses confirmed a reliable reward-driven enhancement that diminished over blocks. The early-block reward effect was most pronounced for participants with stronger color–reward associations, especially during fast-triggered (error-prone) trials. Overall, these results indicate that value-associated distractors can “snap” attention back during lapsing states and prevent commission errors, with some evidence that the potency of this recoupling is tied to the strength of the color–reward association and decreasing with repeated exposure.

Methods

Participants

Using pilot data ($n = 12$; discarded), we performed a power analysis in G*Power (Faul et al., 2007) on RT by accuracy $\times$ distractor feature at 95% power using α criterion of .05. Based on this, we aimed for a sample size of 48 in Experiment 1 and preregistered an analysis plan. A total of 50 participants were recruited from the research participation pool at the University of California, Santa Barbara. Two did not fill out the demographic questionnaire due to time constraints (38 women, 8 men, 2 else; $M \pm SD$; $M_{\text{age}} = 19.54 \text{ years} \pm 1.22$), and most were right-handed (46 right-handed, 1 left-handed, 1 ambidextrous). After analyzing Experiment 1’s full dataset, we recognized that our power analysis was guided by inflated estimates in our small pilot sample. Therefore, for Experiment 2, we selected our intended sample by matching Experiment 1’s sample size. We preregistered an updated analysis plan (see [Analysis](#) subsection) and recruited 50 new participants from the same subject pool. Four did not fill out the questionnaire (31 women, 12 men, 3 else; $M_{\text{age}} = 19.13 \text{ years} \pm 1.15$), and again most were right-handed (45 right-handed, 1 ambidextrous). All participants had normal or corrected color vision in both samples.

Based on our preregistered exclusion criteria, we checked for performance outliers within each sample and found that none were more than 3 standard deviations below the sample mean for accuracy in the training task or A' in the test task. All participants provided informed consent.

Procedure

Participants underwent two phases: a training phase and test phase. In Experiment 1, half of the participants completed these phases one week apart and the other half completed them consecutively (randomly assigned). In Experiment 2, all participants completed them consecutively. All stimuli were presented with Python 3.9.13 and PsychoPy 2023.2.3 (Peirce et al., 2019) and overlaid on a black background. Participants’ viewing distance was 88 cm from a 48×27 -cm monitor. Accuracy was determined by a correct response to a target stimulus and RT was the time in milliseconds (ms) from stimulus presentation until response. During all tasks, white noise was played in the background.

Change-detection task

Upon entering the lab, participants provided informed consent and then completed a working memory task. They completed 12 practice trials followed by 120 trials of a change-detection task across set sizes 4, 6, and 8 (Luck & Vogel, 1997; also used in Anderson et al., 2011 and Milner et al., 2023) to provide a measure of their visual WMC (k). Stimuli ($0.81 \times 0.81^\circ$) were presented in a $13 \times 9.7^\circ$ border with a $0.33 \times 0.33^\circ$ white fixation cross in the center. Cowan’s k was computed for each set size as $S \times (\text{HR} - \text{FA})$, where S is set size, HR is hit rate, and FA is false alarm rate (Cowan, 2001). These values were averaged down to produce a single score per participant.

Training phase

Next, they completed a training task modeled after Experiment 5 in Milner et al. (2023). A sample trial sequence is shown in Fig. 1. On each trial, participants searched a display of six colored circles (1.8° diameter) for a target. The target colors were two colors selected from red [FF0000], blue [0000FF], or green [008000], counterbalanced across participants. Colors of the nontargets were sampled without replacement from cyan [00FFFF], magenta [FF00FF], yellow [FFFF00], gray [D3D3D3], brown [8B4513], orange [FF7F50], and purple [9370DB]. The target circle had a white bar that was either horizontal or vertical; distractor circles had bars that were tilted 45° to the left or right. Participants pressed “m” if the bar (1.3°) in the target ring was horizontal and “z” if it was vertical. Correct responses on trials with one of those target colors were rewarded with \$0.05 75% of the time. The trial ended after 800 ms passed or the participant made a response. Reward feedback was presented afterwards for 1,500 ms. Intertrial intervals were randomly sampled from 500, 600, 700, or 800 ms. Participants completed 48 practice trials with only accuracy feedback. Then, they did 10 blocks of 72 trials each with timed 30-s

TRAINING PHASE

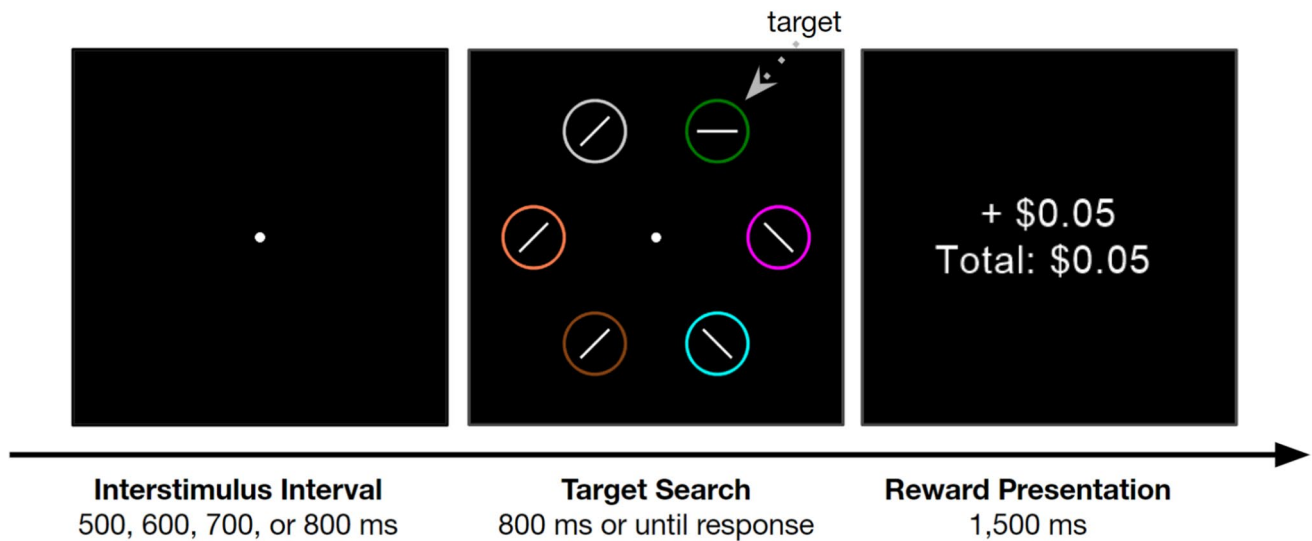


Fig. 1 A schematic of the trial sequence during the training task. After an interstimulus interval, participants saw an array of colored circles. They searched the display for one of two colored circles and

pressed a key based on the line orientation in the target. For one target color (e.g., green), participants were sometimes rewarded if they made the correct response. (Color figure online)

interblock breaks. While most participants completed all 10 blocks, one participant in Experiment 1 did only eight blocks and one participant in Experiment 2 did only six blocks. They were paid the earned monetary reward according to the schedule described above upon completing this phase.

Test phase

Either the week after the training phase or directly afterwards, participants completed a version of the SART with real-time triggering. A sample sequence from three trials of this task is shown in Fig. 2. On each trial, participants searched a display of six shapes for a standout singleton (the circle [1.8° diameter] among squares [1.46 × 1.46°] or the square among circles) and pressed “m” if the bar in that target was horizontal and “z” if it was vertical. Colors were sampled without replacement from the same color list as the nontargets in the training task. One target bar orientation was present for most trials and the other was rare, and the frequent orientation was counterbalanced across participants. Participants had 1,150 ms to respond and there was no interval between trials. The fixation point (0.33° diameter) turned from gray to white when they made a response.

Rare trials were inserted in real time based on intrasubject deviations in RT (i.e., fast or slow RT three-trial windows; below or above 1 standard deviation from the participant’s cumulative mean) to sample optimal and suboptimal attentional states. For instance, for given trial i , if the average RT for $i-3$, $i-2$, and $i-1$ was greater than or less than 1

standard deviation from the average RT up until trial i , trial i would have an infrequent bar orientation in the shape singleton. To avoid tainting the trailing RT window and rule out the effects of recent errors, we required that infrequent trials only be inserted if there had not been any other infrequent trials or errors in the prior three trials. A schematic of this approach is shown in Fig. 3.

Critically, inserted rare trials had a distractor that was either red, blue, green, or neutral. One of these colors was previously rewarded in the training task, allowing us to test value-driven attentional capture. Another color was previously selected in that task, allowing us to test selection-driven attentional capture. The third color served as an unseen, “novel” stimulus feature. The neutral color was taken from the usual color pool and served as a control.

Participants completed 60 practice trials with random rare trials and accuracy feedback presented at the end. They were only allowed to continue if they got around 80% accuracy. They then completed 4 main blocks of 600 trials each with timed 30-s interblock breaks. The first 10% trials of each test block had six randomly inserted infrequent trials and were used to initialize the cumulative RT and standard deviation. The latter 90% was the real-time triggering portion and started off with 100% frequent trials, with up to 54 (27 fast, 27 slow) triggered infrequent ones depending on the RT behavior. While most participants completed all four blocks, one participant in Experiment 1 completed only three; in Experiment 2, two participants completed three blocks, one completed two, and another completed one.

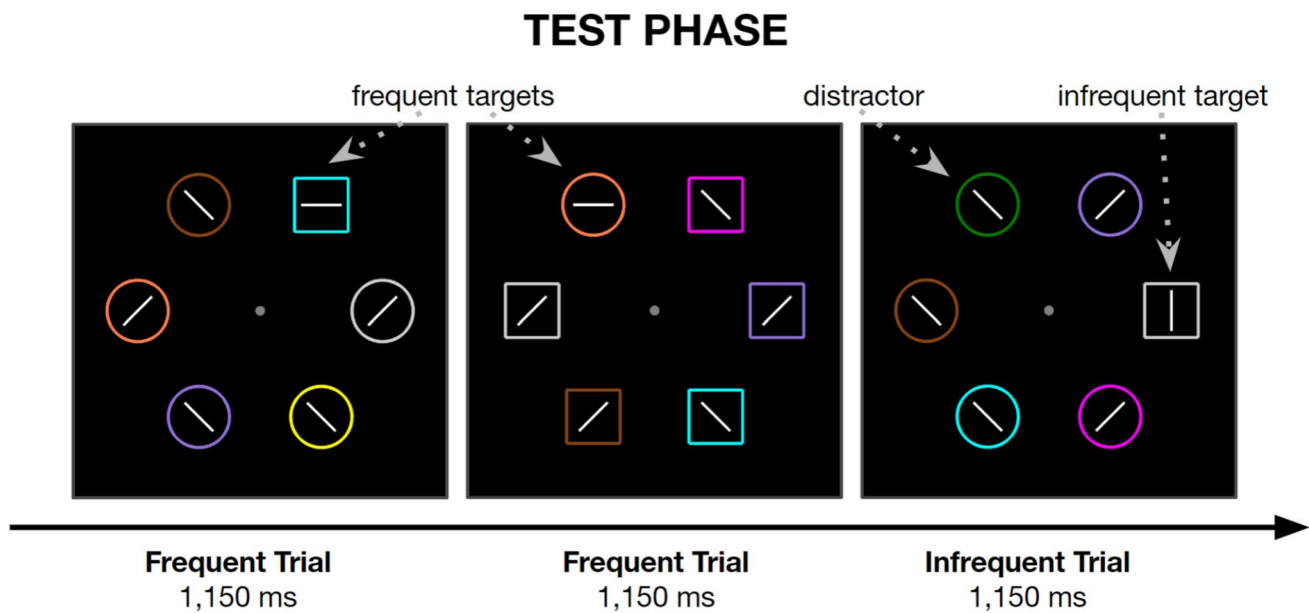


Fig. 2 A sample sequence of three trials from the test phase. Participants searched the display for the shape singleton and made a key press based on the line orientation inside. Frequent trials had targets with one bar orientation inside of them (e.g., horizontal), whereas

infrequent trials had a different orientation (e.g., vertical). Infrequent targets and experience-dependent distractors were inserted in real time based on frequent trial RT speed. (Color figure online)

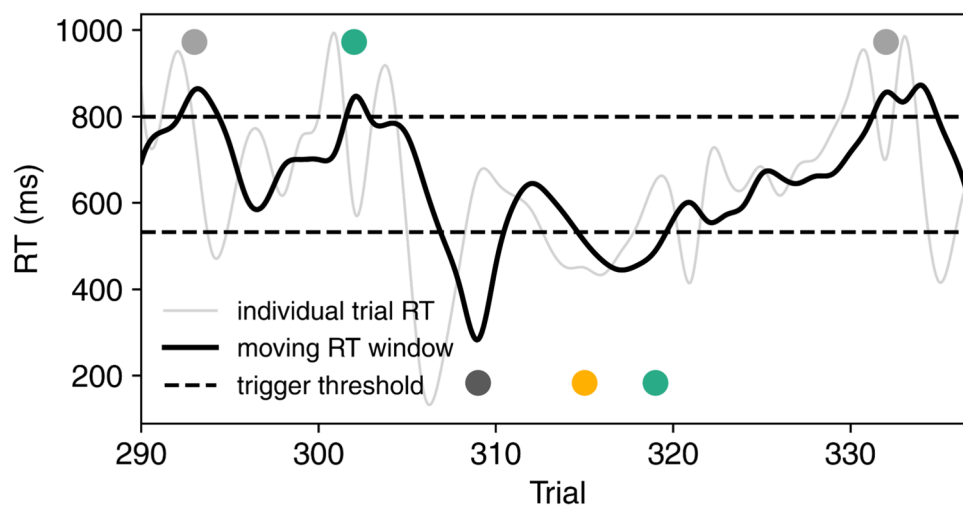


Fig. 3 An example of the triggering procedure for one participant's subset of trials. When the three-trial moving RT average exceeded the upper or lower trigger thresholds, a rare trial was inserted in real

time. The rare trials are the dots, colored by the type of distractor that was present: light gray is neutral, dark gray is selection, gold is novelty, and green is reward. (Color figure online)

Questionnaires

After finishing the test task, participants filled out a questionnaire with demographic and state/trait questions. First, they reported their fatigue compared with the start of their experimental session that day on a 100-point slider scale. They reported their age, gender, and handedness. Next, they filled out the Mind-Wandering Questionnaire (MWQ; Mrazek et al., 2013), a survey on daily mind-wandering.

They also filled out the Mind-Wandering Spontaneous & Deliberate (MWS/MWD; Carriere et al., 2013) scales, which contain items that gauge the frequency of unintentional and intentional task-unrelated thinking during daily life. Finally, they filled out the Effortful Control Scale (ECS; Evans & Rothbart, 2007), which is used to measure trait-level cognitive control (Kawagoe, 2022). One participant did not answer an item in the ECS; we calculated their score using only the completed items.

Analysis

We used R (Version 4.4.2) in R Studio (R Studio Team, 2020) to run all of our data post-processing and statistical analyses. We primarily used mixed-effect models with subject ID assigned as a random effect to account for within-subject variability. This intercept-only structure was chosen a priori as the most parsimonious way to model individual baseline differences. For tests where the dependent variable was RT, we used the `lmer()` function from the *lme4* package and extracted *p* values using the *lmerTest* package. For tests that modeled accuracy, we used the generalized mixed model via the `glmer()` function and specified a binomial distribution with the logit link function. To help the logistic regression models converge, we used the bobyqa optimizer with a maximum of 100,000 function evaluations. We used the *MuMIn* package to calculate marginal and conditional R^2 for our fitted models. We report the unstandardized betas, standard errors, and *p* values (Meteyard & Davies, 2020).

In the test phase attentional capture analysis (i.e., where we asked our primary research questions), we present three assessments for each inferential test: one for Experiment 1 ($n = 50$), one for Experiment 2 ($n = 50$), and a pooled analysis on the combined dataset ($n = 100$). Since both experiments were nearly identical, the first two analyses assess the robustness and reproducibility of the effects while the third (unregistered) leverages increased statistical power for more precise estimates. A generally analogous approach can be

found in Marty-Dugas et al. (2021). Notably, though we considered the perceptual-recoupling account before inspecting Experiment 1's data, our original preregistration (osf.io/58hyw) did not include a defined analysis plan to test recoupling. We therefore conducted those tests on Experiment 1's data as unregistered, exploratory analyses. Based on our findings, we preregistered the recoupling analysis plan for Experiment 2 (osf.io/sev9k) before collecting new data to provide a more rigorous, confirmatory assessment. All analyses from the respective experiments are explicitly labeled in the Results section as preregistered or unregistered (Willroth & Atherton, 2024).

Results

Value-driven learning

To examine reward-driven learning during the training task, we excluded trials with incorrect (Experiment 1: 11.84%; Experiment 2: 12.69%) or no key presses (Experiment 1: 4.74%; Experiment 2: 6.57%) and reaction times (RT) under 200 ms (Experiment 1: 0.06%; Experiment 2: 0.21%). We then performed a linear mixed-effect regression to analyze RT over time by reward condition (see Fig. 4), as preregistered for both experiments. Our model included RT as the dependent variable and trial number and reward condition as an interaction term.

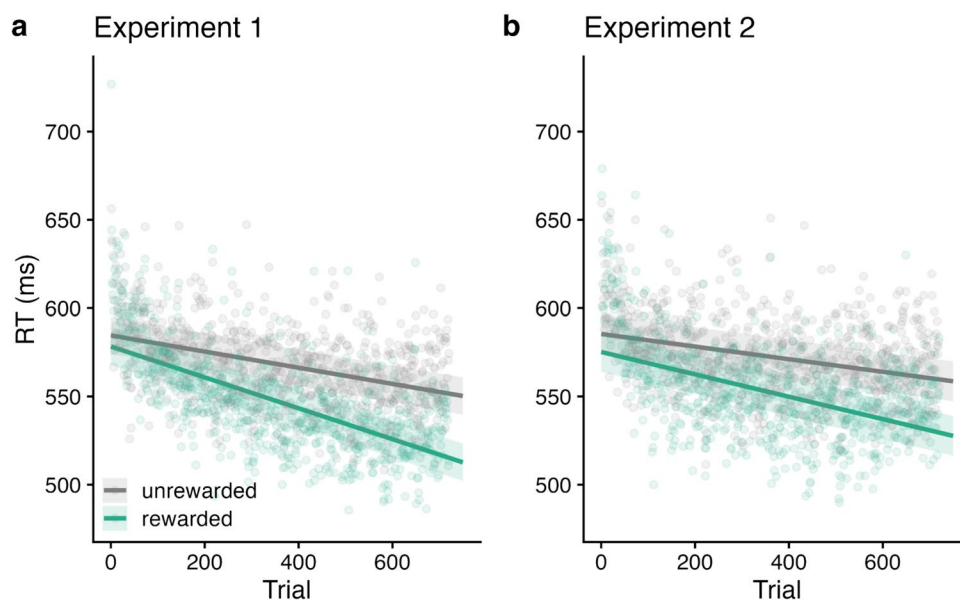


Fig. 4 Linear mixed-effect model predictions of RT over time for accurate unrewarded and rewarded target trials in the training phase of (a) Experiment 1 and (b) Experiment 2. The model was specified with formula $RT \sim \text{reward_condition} \times \text{trial} + (1 \mid \text{subjectID})$. Error shading represents the 95% confidence interval. RTs decreased more

rapidly over time for trials with a rewarded target color compared with unrewarded ones, consistent with value-driven learning. Points depict the mean RT for each trial, computed by averaging across participants within each reward condition. (Color figure online)

In Experiments 1 and 2, RTs were faster on rewarded than unrewarded target trials (Experiment 1: $b = -6.27$, $SE = 1.88$, $p = .00084$; Experiment 2: $b = -10.21$, $SE = 1.98$, $p < .0001$), declined over trials (Experiment 1: $b = -0.046$, $SE = 0.0032$, $p < .0001$; Experiment 2: $b = -0.036$, $SE = 0.0034$, $p < .0001$), and declined more steeply for rewarded target trials (Experiment 1: $b = -0.042$, $SE = 0.0045$, $p < .0001$; Experiment 2: $b = -0.028$, $SE = 0.0048$, $p < .0001$). This confirmed a value-driven learning effect during training.

Triggering procedure validation

To validate the real-time triggering mechanics, we analyzed only triggered rare trials, removing trials with no key presses (Experiment 1: 3.38%; Experiment 2: 4.26%) and RT under 200 ms (Experiment 1: 3.49%; Experiment 2: 4.35%; Mooneyham & Schooler, 2016; Morrison et al., 2014). As expected from the task design, slow triggers followed slower three-trial RT windows

(Experiment 1: $p < .0001$; Experiment 2: $p < .0001$; unregistered). Preregistered mixed-effect logistic regressions confirmed that triggering predicted commission errors (Experiment 1: $b = 1.69$, $SE = 0.060$, $p < .0001$; Experiment 2: $b = 1.83$, $SE = 0.061$, $p < .0001$; Fig. 5). The counts per unique combination of triggers and distractor types (Table 1) for our planned analyses were similar to prior work from our lab and other published work using real-time triggers (Roberts et al., 2025).

No evidence of learned anticipation from distractor color

We assessed whether accuracy changed with block number and its interaction with distractor type to address a concern that participants might learn to predict rare trials based on specific colors (red, blue, green), which would appear as increasing accuracy over time. Accuracy did not change across blocks (Experiment 1: $b = -0.012$,

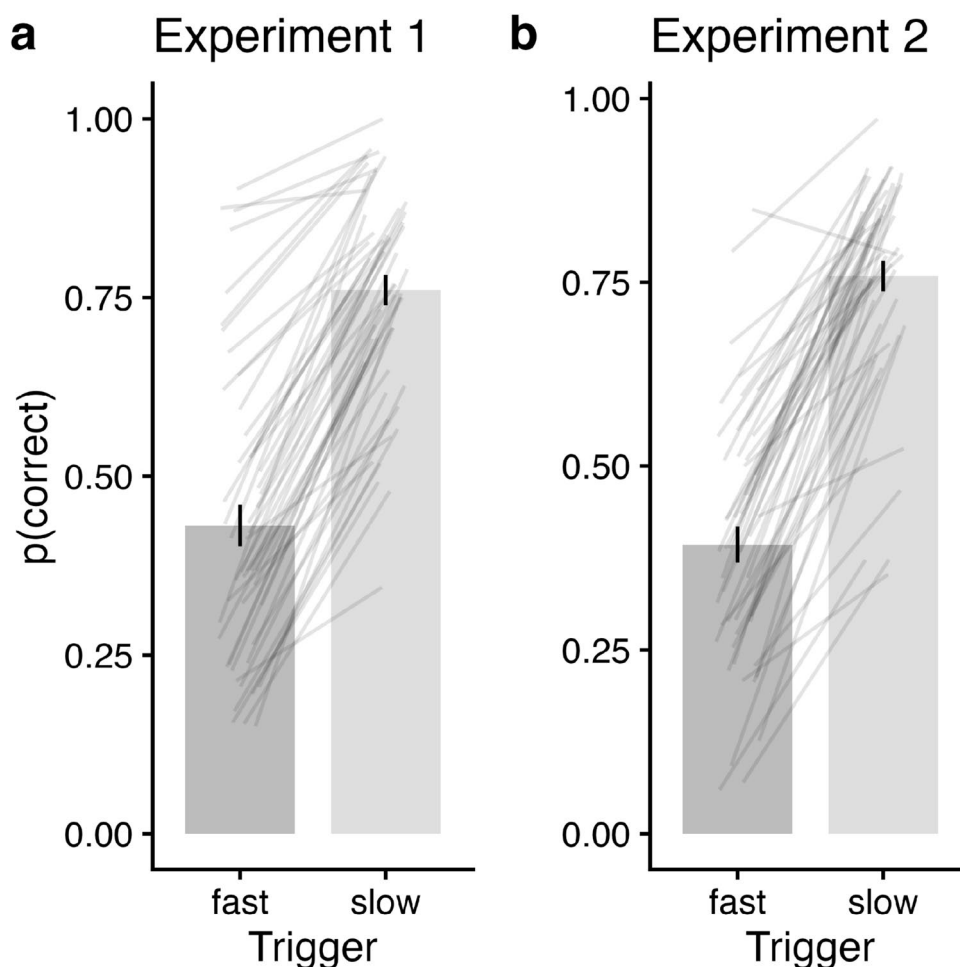


Fig. 5 Mean proportion correct for rare trials that followed fast or slow triggers for (a) Experiment 1 and (b) Experiment 2. Participants were significantly more likely to make errors on fast-triggered trials.

Grey lines connect data points from individual participants. Error bars depict standard error

Table 1 Mean counts and standard deviations for each type of trigger and distractor feature combination across both experiments

Trigger	Distractor feature	Experiment 1 $M \pm SD$	Experiment 2 $M \pm SD$
Fast	Neutral	18.04 $\pm$ 3.78	19.08 $\pm$ 5.81
Fast	Novel	18.34 $\pm$ 5.30	17.28 $\pm$ 4.79
Fast	Reward	18.18 $\pm$ 5.72	18.02 $\pm$ 5.76
Fast	Selection	18.58 $\pm$ 5.51	17.46 $\pm$ 5.18
Slow	Neutral	16.02 $\pm$ 4.54	14.42 $\pm$ 4.78
Slow	Novel	15.56 $\pm$ 4.94	14.36 $\pm$ 4.78
Slow	Reward	15.16 $\pm$ 4.86	13.68 $\pm$ 4.25
Slow	Selection	15.74 $\pm$ 5.48	14.64 $\pm$ 5.10

$SE = 0.024$, $p = .62$, preregistered; Experiment 2: $b = -0.0010$, $SE = 0.024$, $p = .97$, unregistered; pooled: $b = -0.0068$, $SE = 0.017$, $p = .69$). When including the interaction with distractor type, reward-associated distractors significantly boosted accuracy over neutral ones in Experiment 1 ($b = 0.39$, $SE = 0.19$, $p = .037$; preregistered). The reward distractor–block interaction was nonsignificant ($b = -0.12$, $SE = 0.067$, $p = .066$), though its direction suggested a decreasing benefit over blocks. These effects did not replicate in Experiment 2 (unregistered): reward-associated distractors neither influenced accuracy ($b = 0.064$, $SE = 0.18$,

$p = .73$) nor interacted with block number ($b = 0.031$, $SE = 0.067$, $p = .65$). Pooled results showed no main effect of reward ($b = 0.23$, $SE = 0.13$, $p = .083$) and no interaction between reward and block ($b = -0.047$, $SE = 0.047$, $p = .32$). Crucially, none of the models showed rising accuracy across blocks for any distractor, ruling out color-based learning.

Perceptual recoupling

Experiment 1's early accuracy boost from reward-driven distractors aligned with predictions from the perceptual-recoupling account (see Appendix A for an exploratory check on the durability of this effect across a weeklong delay). Adding a three-way interaction with the trigger condition to test this effect across attentional states (Fig. 6), we found (unregistered) that both novelty and reward-driven distractors enhanced accuracy ($b = 0.54$, $SE = 0.26$, $p = .039$; $b = 0.75$, $SE = 0.26$, $p = .0032$) with no difference across fast/slow triggers. The reward effect again declined over blocks ($b = -0.23$, $SE = 0.094$, $p = .016$). In Experiment 2 (preregistered), there were no novelty ($b = 0.082$, $SE = 0.24$, $p = .73$) or reward effects ($b = 0.41$, $SE = 0.24$, $p = .089$), but there was a significant interaction, such that the reward effect was stronger during fast-triggered than slow-triggered trials ($b = -0.87$, $SE = 0.43$, $p = .045$). A three-way interaction with block number was nonsignificant ($b = 0.28$, $SE = 0.16$,

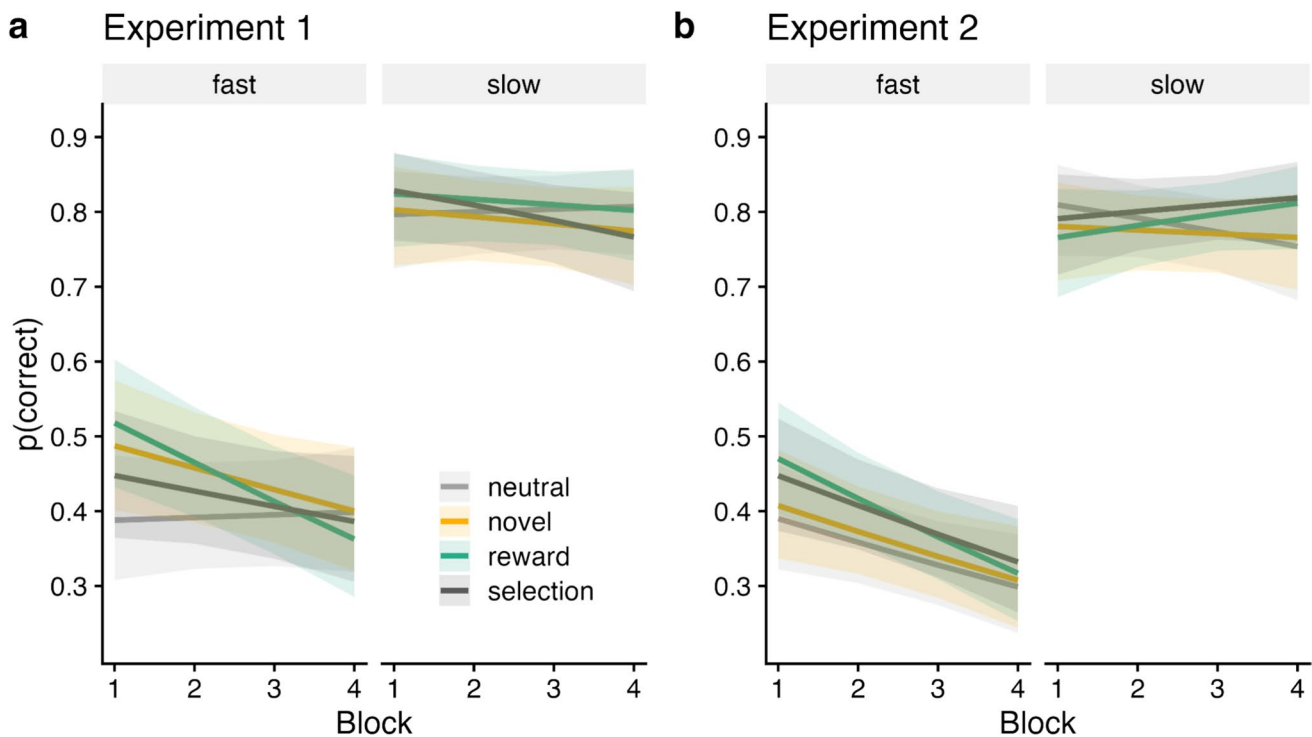


Fig. 6 Mixed-effect logistic regression model predictions for the probability of a correct response as a function of block, distractor type, and trigger condition for (a) Experiment 1 and (b) Experiment

2. The model formula was accuracy $\sim$ block $\times$ trigger $\times$ distractor + (1 | subjectID). The shading around each line represents the 95% confidence interval. (Color figure online)

$p=.069$), though its direction was consistent with a pattern in which reward-associated distractors boosted accuracy during fast-triggered trials in the early blocks of Experiment 2.

To recap, Experiment 1 showed that reward-associated distractors enhanced accuracy more in early blocks, while Experiment 2 showed greater enhancement during fast-triggered trials. These findings confirm that value-driven capture reduced commission errors, though the timing varied across experiments. Pooling the samples together for greater power (Table 2), reward-associated distractors significantly boosted accuracy ($b=0.57$, $SE=0.17$, $p=.0011$). This effect declined over blocks ($b=-0.15$, $SE=0.065$, $p=.020$) and was stronger during fast-triggered trials ($b=-0.65$, $SE=0.30$, $p=.030$). The three-way interaction was not significant, but the pattern was that reward-associated distractors particularly enhanced accuracy during fast-triggered trials in early blocks ($b=0.21$, $SE=0.11$, $p=.053$).

We next tested whether individual differences influenced capture during certain attentional states. Value-driven capture enhanced accuracy across attentional states but weakened over blocks; therefore, this analysis focused on the first two blocks (Fig. 7), where the capture effect was strongest. One candidate moderator is “reward learning magnitude,” calculated as the RT difference between accurate rewarded and unrewarded target trials during training. Higher values indicated a stronger color–reward association. In Experiment 1 (unregistered), a two-way interaction showed that reward-associated distractors improved accuracy for

individuals with a high learning magnitude ($b=0.018$, $SE=0.0079$, $p=.020$). Two three-way interactions revealed that novelty-associated and reward-associated distractors boosted accuracy during fast triggers for high learners ($b=-0.030$, $SE=0.013$, $p=.021$; $b=-0.034$, $SE=0.013$, $p=.0090$). None of these effects were detected in Experiment 2 (preregistered); no interaction between reward-driven distractors and learning magnitude ($b=0.0072$, $SE=0.0054$, $p=.18$), and no three-way interactions with novelty-driven ($b=-0.0092$, $SE=0.0094$, $p=.33$) and reward-driven capture ($b=-0.0097$, $SE=0.0097$, $p=.32$) during specific trigger types.

Our pooled model failed to converge despite doubling bobyqa iterations and testing alternative optimizers. We report the resulting estimates with appropriate caution, noting that they should be interpreted as only suggestive rather than conclusive due to the convergence failure (Table 3). The two-way interaction between distractor feature and learning magnitude was significant only for reward-associated distractors ($b=0.011$, $SE=0.0045$, $p=.017$), indicating greater accuracy benefits for participants with a stronger color–reward association. Three-way interactions with learning magnitude and trigger type were significant for novel and reward distractors ($b=-0.016$, $SE=0.0075$, $p=.031$; $b=-0.018$, $SE=0.0077$, $p=.021$). These negative three-way coefficients imply that accuracy benefit from both distractor types was greater during fast-triggered trials for high learners.

Table 2 Mixed-effect logistic regression model results when fitting accuracy as a function of block, trigger, and distractor type on pooled data from both experiments

Term	<i>b</i>	<i>SE</i>	95% CI	<i>z</i>	<i>p</i>
(Intercept)	−0.38	0.15	[−0.67, −0.090]	−2.56	.010
Block	−0.063	0.046	[−0.15, 0.027]	−1.37	.17
Distractor (novel)	0.29	0.18	[−0.055, 0.63]	1.64	.10
Distractor (reward)	0.57	0.17	[0.23, 0.91]	3.27	.0011
Distractor (selection)	0.29	0.18	[−0.052, 0.64]	1.67	.095
Trigger (slow)	1.82	0.21	[1.41, 2.23]	8.67	<.0001
Block:distractor (novel)	−0.064	0.064	[−0.19, 0.064]	−0.98	.33
Block:distractor (reward)	−0.15	0.065	[−0.28, −0.024]	−2.32	.020
Block:distractor (selection)	−0.061	0.066	[−0.19, 0.068]	−0.92	.36
Block:trigger (slow)	0.026	0.075	[−0.12, 0.17]	0.34	.73
Distractor (novel):trigger (slow)	−0.34	0.30	[−0.92, 0.24]	−1.14	.25
Distractor (reward):trigger (slow)	−0.65	0.30	[−1.25, −0.062]	−2.16	.030
Distractor (selection):trigger (slow)	−0.22	0.30	[−0.81, 0.37]	−0.73	.46
Block:distractor (novel):trigger (slow)	0.056	0.11	[−0.15, 0.26]	0.53	.60
Block:distractor (reward):trigger (slow)	0.21	0.11	[−0.0029, 0.42]	1.93	.053
Block:distractor (selection):trigger (slow)	0.054	0.11	[−0.16, 0.26]	0.50	.62
Marginal R^2	.17				
Conditional R^2	.31				

Note: Bold text highlights rows where $p < .05$

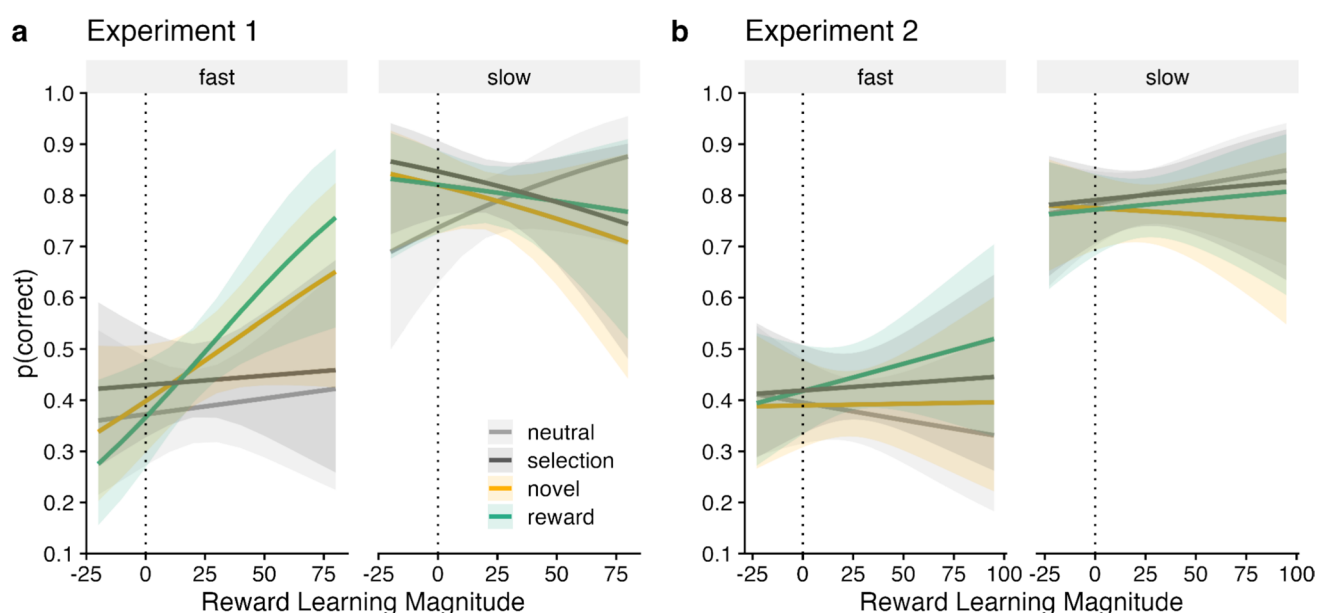


Fig. 7 Mixed-effect logistic regression model predictions for (a) Experiment 1 and (b) Experiment 2's probability of a correct response in the first two blocks as a function of distractor type, trigger condition, and learning magnitude from the training task. The model formula was $\text{accuracy} \sim \text{trigger} \times \text{distractor} \times \text{learning_magnitude} + (1 | \text{subjectID})$. The shading around each line represents the 95% confidence interval. To the right of the vertical dotted line is where participants were faster for rewarded targets than unrewarded targets during training. (Color figure online)

subjectID). The shading around each line represents the 95% confidence interval. To the right of the vertical dotted line is where participants were faster for rewarded targets than unrewarded targets during training. (Color figure online)

Table 3 Mixed-effect logistic regression model results when fitting accuracy as a function of trigger, distractor type, and learning magnitude on pooled data from both experiments

Term	<i>b</i>	<i>SE</i>	95% CI	<i>z</i>	<i>p</i>
(Intercept)	−0.45	0.14	[−0.73, −0.17]	−3.16	.0016
Trigger (slow)	1.62	0.17	[1.30, 1.95]	9.74	<.0001
Distractor (novel)	0.049	0.14	[−0.23, 0.33]	0.34	.73
Distractor (reward)	0.072	0.14	[−0.21, 0.35]	0.50	.62
Distractor (selection)	0.15	0.14	[−0.13, 0.43]	1.07	.28
Learning magnitude	−0.00095	0.0045	[−0.0097, 0.0078]	−0.21	.83
Trigger (slow):distractor (novel)	0.12	0.24	[−0.34, 0.58]	0.52	.60
Trigger (slow):distractor (reward)	0.11	0.24	[−0.37, 0.59]	0.45	.65
Trigger (slow):distractor (selection)	0.15	0.24	[−0.32, 0.63]	0.62	.53
Trigger (slow):learning magnitude	0.0081	0.0055	[−0.0026, 0.019]	1.48	.14
Distractor (novel):learning magnitude	0.0061	0.0045	[−0.0027, 0.015]	1.36	.17
Distractor (reward):learning magnitude	0.011	0.0045	[0.0019, 0.019]	2.39	.017
Distractor (selection):learning magnitude	0.0021	0.0044	[−0.0065, 0.011]	0.48	.63
Trigger (slow):distractor (novel):learning magnitude	−0.016	0.0075	[−0.031, −0.0014]	−2.15	.031
Trigger (slow):distractor (reward):learning magnitude	−0.018	0.0077	[−0.033, −0.0027]	−2.31	.021
Trigger (slow):distractor (selection):learning magnitude	−0.010	0.0077	[−0.025, 0.0051]	−1.30	.19
Marginal R^2	.15				
Conditional R^2	.27				

Note: Bold text highlights rows where $p < .05$

As an additional check, we tested whether distractors produced RT slowing, a standard index of attentional capture. In an unregistered analysis suggested by a reviewer,

we modeled RT as a function of distractor type and trigger (Fig. 8) on the combined dataset. Reward-associated distractors significantly slowed RTs on the rare trials ($b = 12.29$,

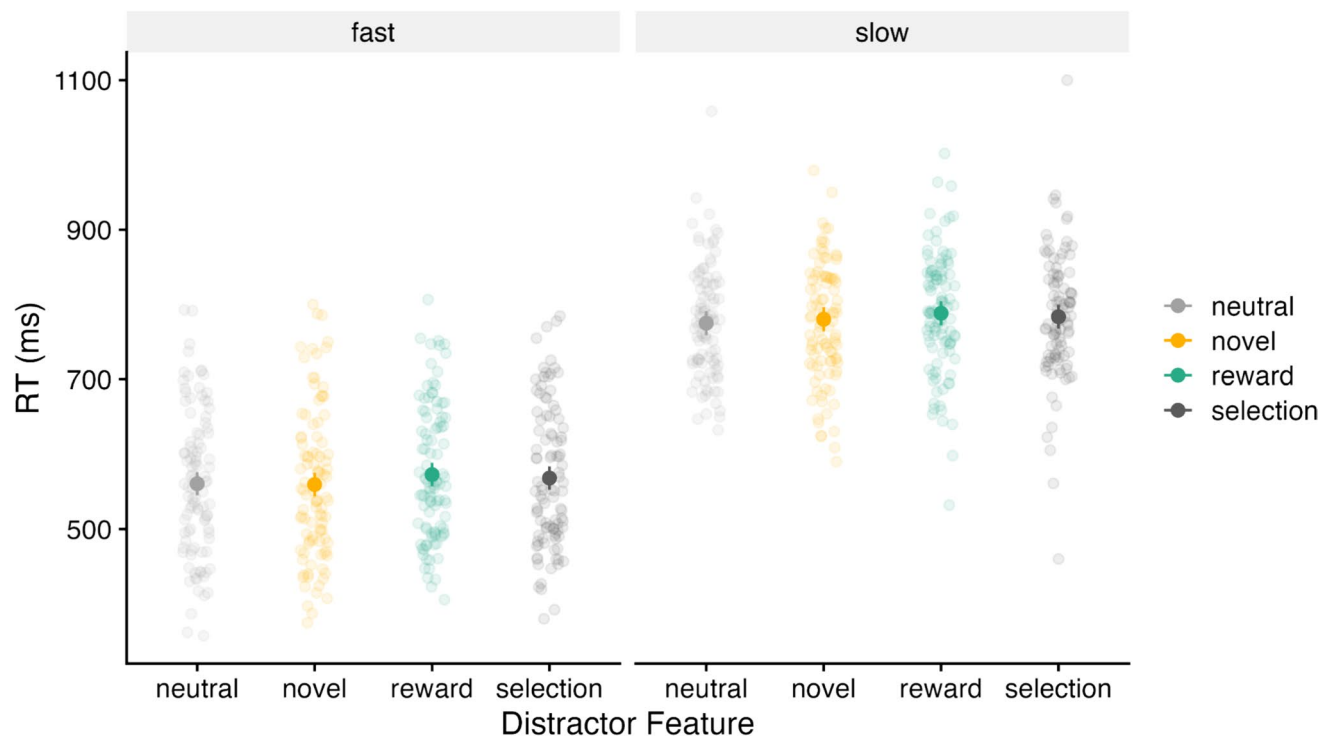


Fig. 8 Linear mixed-effect model predictions of RT by trigger and distractor in combined test data. The model formula was $RT \sim \text{distractor} \times \text{trigger} + (1 \mid \text{subjectID})$. Jittered points show per-participant RT

averages per unique combination of distractor type and trigger. Error bars depict the 95% confidence interval. (Color figure online)

$SE = 5.78$, $p = .033$). There were no RT effects for novelty ($p = .86$) or selection ($p = .19$), nor any interactions with trigger type (all p values $> .47$). Thus, the general pattern in the results suggested that reward-associated distractors both slowed RTs and improved accuracy on rare trials, reinforcing that value-driven attentional capture occurred. See Appendix B for additional analyses confirming evidence of capture across triggers, Appendix C for robustness checks on our main test phase analyses using an iterative maximal-to-parsimonious random effects selection procedure, and Appendix D for a simulation-based power check for the main test phase analyses.

Capture suppression

The results so far corroborate the predictions of the perceptual-recoupling account. Recall that the capture-suppression account predicts that distractors might slow rare trial RTs during errors due to an unavailability of attention resources to suppress capture. We performed planned analyses on data from all four blocks to assess the capture-suppression account. When fitting a linear mixed-effect model with RT as the dependent variable and the interaction between accuracy and distractor type as the predictor, there were no effects of distractors nor interactions with accuracy in Experiment 1 (preregistered; all p values $> .15$),

Experiment 2 (unregistered; all p values $> .17$), or the pooled analysis (all p values $> .12$). Next, k was added to the model to assess the effect on RT associated with WMC. In Experiment 1 (preregistered), trials with a selection-associated distractor were slower than those with a neutral distractor ($b = 62.28$, $SE = 24.28$, $p = .010$) and participants with higher k were less susceptible to this slowdown ($b = -25.26$, $SE = 10.50$, $p = .016$). This did not replicate in Experiment 2 (unregistered; all p values $> .23$) or the pooled analysis (all p values $> .20$).

Given that the previous recoupling analysis suggested that RT was slowed by some distractors, how is it the case that our planned capture-suppression tests here showed inconsistent evidence of RT slowing? The discrepancy is because the two analyses are based on different variables and are designed to ask different questions. The capture-suppression analyses were based on rare trial accuracy (error vs. correct) and asked whether distractor-related RT slowing was larger on trials that ended in error. By contrast, recoupling analyses were based on trigger type, and tested whether distractors slowed RTs during lapse-triggered moments regardless of whether they ended in an error or not. Because distractors themselves often prevented errors, sorting trials by accuracy likely obscured RT slowing that happened during lapses. These preregistered capture-suppression analyses were therefore not a good fit for the data.

Discussion

Lapses of sustained attention are frequent in everyday life and can be deeply consequential. Identifying how lapses affect other domains of cognitive processing has been the aim of recent work that leverages a closed-loop triggering procedure to isolate moments of optimal and suboptimal attention (for a review, see Shelat et al., 2024). In the current study, we assessed how experience-dependent capture by a distractor affected behavioral performance across attentional states in a continuous response task (SART) with real-time triggering. We tested predictions from two accounts: one predicted that capture suppression should be impaired during lapses due to failures of cognitive control, which would emerge as distractor capture interfering with SART performance (i.e., the capture-suppression account); the other predicted that capture, even when from a task-irrelevant distractor, may recouple attention and perception during lapsing states to prevent errors (i.e., the perceptual-recoupling account). Experiment 1 showed that both novelty-associated and value-associated distractors enhanced accuracy, whereas Experiment 2 showed a moderation of the value-driven effect during fast-triggered moments. Pooled analyses confirmed a reliable value-driven enhancement but did not consistently support novelty-driven capture. Critically, the value–accuracy effect was most pronounced early in the task, suggesting an “extinction” pattern where participants plausibly unlearned the color–reward association over blocks (Milner et al., 2023). Altogether, the results favored predictions from the perceptual-recoupling account.

It is important to note that the capture-suppression and perceptual-recoupling accounts are not mutually exclusive, and rather may describe different stages of the same process. A lapse of attention may initially impair goal-directed suppression, allowing a distractor to capture attention. That same capture event could then re-engage perceptual processing. In this sense, reduced suppression could allow capture, and the resulting perceptual recoupling could repair the lapse. Our behavioral data therefore suggest that the latter stage dominates performance in the SART with real-time triggering. This also raises the question of whether value-driven capture is unique, or whether any manipulation that induces capture and RT slowing could promote perceptual recoupling. Here, it appears that reward history is one effective route, though recoupling itself may be a more general reflection of attentional capture during lapses (Smallwood, 2013).

One caveat is that our SART with visual search on each trial is likely not an adequate way to measure capture suppression. That is, errors on this task may be mostly due to sustained attention failures rather than failures of the attentional processes that support visual search (cf. deBettencourt et al., 2021; Smallwood et al., 2008a).

Consequently, the results should be interpreted with caution when generalizing to contexts where performance is more contingent on spatial attention. Future research should consider using a task specifically designed to tease apart the effects of capture suppression from those of sustained attention to better assess the underlying mechanisms of attentional control. This would clarify the boundaries of the perceptual-recoupling hypothesis and also refine our understanding of how attentional capture is modulated in different cognitive tasks.

A second concern is the potential confound of a speed–accuracy tradeoff shaping RT variability and accuracy in our task. Indeed, much work has explored alternative explanations for commission errors that have nothing to do with failures of sustained attention (e.g., Seli et al., 2013; Seli et al., 2012). However, for this to be a viable account of our findings, a significant strategy shift would have to occur within the window of a single rare trial, which is unlikely. A more likely explanation is some sort of “snap back into it” effect driven by perceptual recoupling, enabling the correct response. Future studies could aim to replicate the behavioral findings identified here while tracking other correlates of attentional state (e.g., event-related potentials, pupil diameter) to rule out the speed–accuracy trade-off concern and isolate the perceptual mechanism that drove our reduction in commission errors during fast triggers. It is also important to test whether these effects generalize to sustained attention paradigms that do not have a speed–accuracy trade-off confound (see Skinner & Giesbrecht, 2025).

The core pattern of results from both experiments is that value-driven attentional capture reduces commission errors. This aligns with Smallwood (2013), where physical salience reduced SART errors by recoupling attention with external perceptual input. Our findings extend this by showing that some types of *experience* (i.e., prior reward history with color) can shape attentional priority. More broadly, this opens the door to using personalized, reward-based training and adaptive interventions that harness previous experiences to improve performance in high-stakes environments.

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Data availability All data, including pilot data, are available in the OSF repository.

Code availability All code is available in the OSF repository.

Declarations

Conflicts of interest/Competing interests The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Ethics approval The procedures were approved by the University of California, Santa Barbara Human Subjects Committee and the Army Research Office/Human Research Protection Office.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication The authors affirm that human research participants provided informed consent for publication. There is no identifying information of individual participants in this article.

Open practices statement Experiment 1 was fully preregistered at osf.io/58hyw. Experiment 2 was fully preregistered at osf.io/sev9k. All data (including pilot data) and materials are available at osf.io/anwzx.

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